Homology of the Torus

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of homology of the torus, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AT-03 / Algebraic Topology / Report v1

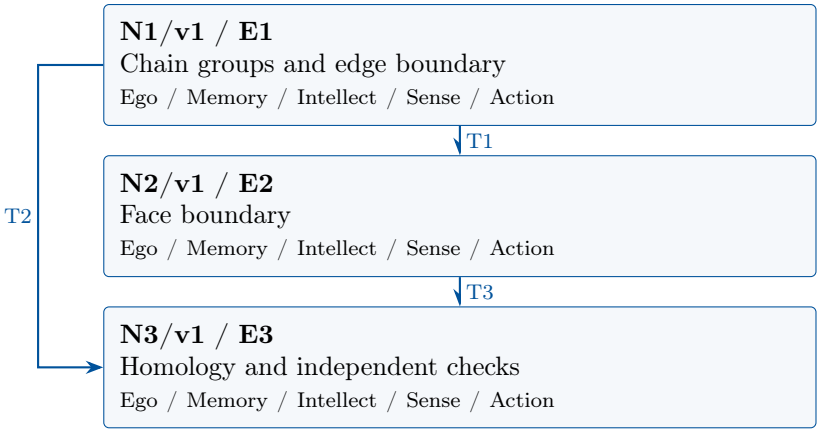
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Compute integral homology of T^2: $H_0 \cong \mathbb{Z}$, $H_1 \cong \mathbb{Z}^2$, $H_2 \cong \mathbb{Z}$, and $H_k = 0$ for $k > 2$.*

A. Foundations and permitted imports

Use the square quotient CW structure on T^2: one vertexv, oriented edgesa,b, and one2-cellc attached alongaba $^{-1}b^{-1}$. Import cellular homology with integral coefficients: C_k is free on k-cells, and its differential is the attaching-map incidence degree after collapsing the other cells. Cellular homology agrees with singular homology. AT-02 supplies this model; its π_1 result is not required for the boundary computation.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The chain groups and zero degree 1 differential.

Receiving activity. N2: Face boundary

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Attaching wordaba ${}^{-1}b^{-1}$; prior exports E1/v1

T2: N1/E1/v1 to N3. The chain groups and zero degree 1 differential.

Receiving activity. N3: Homology and independent checks

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Both zero differentials; prior exports E1/v1, E2/v1

T3: N2/E2/v1 to N3. The degree 2 differential is zero.

Receiving activity. N3: Homology and independent checks

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Both zero differentials; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Chain groups and edge boundary

Object and role. Count cells and compute endpoint incidences.

Inputs. The CW cell model; foundational imports stated in §1

E — Ego

Own E1/v1: The chain groups and zero degree 1 differential.

M — Memory

Retain The CW cell model; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Count cells and compute endpoint incidences.

S — Sense

Identified endpoints still retain their opposite incidence signs.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 The cellular chain groups are $C_2 = Z\langle c\rangle$, $C_1 = Z\langle a, b\rangle$, $C_0 = Z\langle v\rangle$ and zero in higher degrees.

N1.2 Both endpoints of each edge arev, with opposite endpoint signs, so $\partial_1 a = v - v = 0$ and $\partial_1 b = 0$.

Export E1/v1. The chain groups and zero degree 1 differential.

Acceptance gate. Identified endpoints still retain their opposite incidence signs. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Identified endpoints still retain their opposite incidence signs. Recheck dependent nodes N2, N3.

B. N2/v1: Face boundary

Object and role. Compute signed degree separately for each target edge.

Inputs. Attaching wordaba $^{-1}b^{-1}$; prior exports E1/v1

E — Ego

Own E2/v1: The degree 2 differential is zero.

M — Memory

Retain Attaching wordaba $^{-1}b^{-1}$; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

Compute signed degree separately for each target edge.

S — Sense

Counting appearances without orientation would incorrectly produce2a+2b.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Collapse all 1-cells excepta when computing the coefficient ofa in $\partial_2 c$.

N2.2 The attaching loop crossesa once forward and once backward, giving degree $1 - 1 = 0$.

N2.3 The same calculation for b gives 0.

N2.4 Therefore $\partial_2 c = 0a + 0b$.

N2.5 This is a signed incidence computation, not an assumption that every closed surface has zero cellular boundary in every chosen decomposition.

Export E2/v1. The degree 2 differential is zero.

Acceptance gate. Counting appearances without orientation would incorrectly produce2a+2b. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Counting appearances without orientation would incorrectly produce2a+2b. Recheck dependent nodes N3.

C. N3/v1: Homology and independent checks

Object and role. Compute kernels modulo images in each degree.

Inputs. Both zero differentials; prior exports E1/v1, E2/v1

E — Ego

Own E3/v1: All integral homology groups in the target.

M — Memory

Retain Both zero differentials; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

I — Intellect

Compute kernels modulo images in each degree.

S — Sense

Euler characteristic alone cannot determine every homology group or its torsion.

A — Action

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 The chain complex is $0 \to Z \to^0 Z^2 \to^0 Z \to 0$, so $H_2 = \ker \partial_2 = Z$, $H_1 = \ker \partial_1 / \operatorname{im} \partial_2 = Z^2$, $H_0 = C_0 / \operatorname{im} \partial_1 = Z$, and $H_k = 0$ for $k > 2$.

N3.2 Its alternating rank sum $1 - 2 + 1 = 0$ matches the cell count.

N3.3 If independently using AT-02, H_1 matches the abelianization of $\pi_1(T^2) = Z^2$.

N3.4 These checks support consistency but the kernel/image computation establishes the groups.

Export E3/v1. All integral homology groups in the target.

Acceptance gate. Euler characteristic alone cannot determine every homology group or its torsion. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Euler characteristic alone cannot determine every homology group or its torsion. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Chain groups and edge boundary

1. The cellular chain groups are $C_2 = Z\langle c\rangle$, $C_1 = Z\langle a, b\rangle$, $C_0 = Z\langle v\rangle$ and zero in higher degrees.
2. Both endpoints of each edge arev, with opposite endpoint signs, so $\partial_1 a = v - v = 0$ and $\partial_1 b = 0$.

B. §4.N2 — Face boundary

1. Collapse all 1-cells excepta when computing the coefficient ofa in $\partial_2 c$.
2. The attaching loop crossesa once forward and once backward, giving degree $1 - 1 = 0$.
3. The same calculation for b gives 0.
4. Therefore $\partial_2 c = 0a + 0b$.
5. This is a signed incidence computation, not an assumption that every closed surface has zero cellular boundary in every chosen decomposition.

C. §4.N3 — Homology and independent checks

1. The chain complex is $0 \to Z \to^0 Z^2 \to^0 Z \to 0$, so $H_2 = \ker \partial_2 = Z$, $H_1 = \ker \partial_1 / \operatorname{im} \partial_2 = Z^2$, $H_0 = C_0 / \operatorname{im} \partial_1 = Z$, and $H_k = 0$ for $k > 2$.
2. Its alternating rank sum $1 - 2 + 1 = 0$ matches the cell count.
3. If independently using AT-02, H_1 matches the abelianization of $\pi_1(T^2) = Z^2$.
4. These checks support consistency but the kernel/image computation establishes the groups.

□

V. WORKED ORCHESTRATION

The oriented2-cellc is a generator of H_2 because its boundary is 0 and there are no3-cells to make it a boundary. The classesa,b independently generateH $_1$; an integer $cycle3a - 2b$ is nonzero because $\operatorname{im} \partial_2 = 0$.

A. Boundary and hypothesis audit

For RP^2 the 2-cell attaches by a degree 2 loop, so ∂_2 is multiplication by 2 and $H_1 = Z/2$, $H_2 = 0$. Sense uses this comparison to reject the claim that all face boundaries vanish merely because an attaching word is closed.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For RP^2 the 2-cell attaches by a degree 2 loop, so ∂_2 is multiplication by 2 and $H_1 = Z/2$, $H_2 = 0$. Sense uses this comparison to reject the claim that all face boundaries vanish merely because an attaching word is closed.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Allen Hatcher, Algebraic Topology: covering spaces, cellular homology, and cap products.